

Chebyshev's Inequality :

Statement: If X is a random variable with mean μ and variance σ^2 , then for any positive number K , we have

$$P\{|X - \mu| \geq K\sigma\} \leq \frac{1}{K^2} \quad \text{--- (*)}$$

$$\text{or } P\{|X - \mu| < K\sigma\} \geq 1 - \frac{1}{K^2} \quad \text{--- (**)}$$

Remark: In particular, if we take $K\sigma = C > 0$, then (*) and (**) give respectively

$$P\{|X - \mu| \geq C\} \leq \frac{\sigma^2}{C^2} \quad \text{and} \quad P\{|X - \mu| < C\} \geq 1 - \frac{\sigma^2}{C^2}$$

$$\Rightarrow P\{|X - E(X)| \geq C\} \leq \frac{\text{Var}(X)}{C^2}$$

$$\text{and } P\{|X - E(X)| < C\} \geq 1 - \frac{\text{Var}(X)}{C^2}$$

Generalised form of Bienayme-Chebyshev's Inequality:

Let $g(X)$ be a non-negative function of a random variable X , then for every $K > 0$, we have

$$P\{g(X) \geq K\} \leq \frac{E\{g(X)\}}{K} \quad \text{--- (***)}$$

Remark 1: If we take $g(X) = \{X - E(X)\}^2 = \{X - \mu\}^2$ and replace K by $K^2\sigma^2$ in (***) we get

$$P\{(X - \mu)^2 \geq K^2\sigma^2\} \leq \frac{E(X - \mu)^2}{K^2\sigma^2} = \frac{\sigma^2}{K^2\sigma^2} = \frac{1}{K^2}$$

$$\Rightarrow P\{|X - \mu| \geq K\sigma\} \leq \frac{1}{K^2}$$

which is Chebyshev's inequality.

2. Markov's Inequality: If we take $g(X) = |X|$, we get, for any $K > 0$

$$P\{|X| \geq K\} \leq \frac{E|X|}{K}, \text{ which is Markov's inequality.}$$

Rather taking $g(x) = |x|^r$ and replacing K by K^r , we get a more general form of Markov's inequality, viz,

$$P(|X|^r \geq K^r) \leq \frac{E|X|^r}{K^r}.$$

Convergence in Probability:

The convergence in probability or stochastic convergence which is defined as follows:

Defⁿ: A sequence of random variables $X_1, X_2, \dots, X_n$ is said to converge in probability to a constant a , if for any $\epsilon > 0$,

$$\lim_{n \rightarrow \infty} P(|X_n - a| < \epsilon) = 1$$

$$\text{or } \lim_{n \rightarrow \infty} P(|X_n - a| \geq \epsilon) = 0$$

and write $X_n \xrightarrow{P} a$, as $n \rightarrow \infty$.

If there exists a random variable X such that $X_n - X \xrightarrow{P} 0$ as $n \rightarrow \infty$, then we say that the given sequence $\{X_n\}$ of random variables converges in probability to the r.v. X .

Remark: If $X_n \xrightarrow{P} \alpha$ and $Y_n \xrightarrow{P} \beta$ as $n \rightarrow \infty$, then

$$(i) \quad X_n \pm Y_n \xrightarrow{P} \alpha \pm \beta \text{ as } n \rightarrow \infty$$

$$(ii) \quad X_n Y_n \xrightarrow{P} \alpha \beta \text{ as } n \rightarrow \infty$$

$$(iii) \quad \frac{X_n}{Y_n} \xrightarrow{P} \frac{\alpha}{\beta} \text{ as } n \rightarrow \infty, \text{ provided } \beta \neq 0.$$

Chebychev's Theorem: As an immediate consequence of Chebychev's inequality, we have the following theorem on convergence in probability.

" If $X_1, X_2, \dots, X_n$ is a sequence of random variables and if mean μ_n and standard deviation σ_n of X_n exists for all n and if $\sigma_n \rightarrow 0$ as $n \rightarrow \infty$, then $X_n - \mu_n \xrightarrow{P} 0$ as $n \rightarrow \infty$.

Proof: By Chebychev's inequality, for $\epsilon > 0$,

$$P\{|X_n - \mu_n| \geq \epsilon\} \leq \frac{\sigma_n^2}{\epsilon^2} \rightarrow 0 \text{ as } n \rightarrow \infty \quad (\because \sigma_n \rightarrow 0 \text{ as } n \rightarrow \infty)$$

Hence $X_n - \mu_n \xrightarrow{P} 0$ as $n \rightarrow \infty$, provided $\sigma_n \rightarrow 0$ as $n \rightarrow \infty$.

LAW of Large Numbers (LLN)

The basic idea behind the LLN is that when a physical quantity is measured the measurements are subjected to random errors, the true measurements of the quantity can be approximated by arithmetic mean of the measurements. And the accuracy of the approximation increases with the number of measurements, hence the name LLN. Here, the basic assumption is that both positive and negative errors will occur in the measurements and when averaged they will cancel each other to a large extent; it is very unlikely that errors will accumulate in one direction as the number of measurements increases.

Symbolically - let $\{X_i\}, i=1, 2, \dots, n$, be a sequence of measurements and let $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$, be the mean of n measurements i.e. $\{\bar{X}_n\}, n=1, 2, \dots$ is also a sequence. In LLN we study the conditions under which $\bar{X}_n \rightarrow \mu$, μ is the true value, by different methods of convergence.

Different LLN are obtained by changing two factors viz (1) the type of convergence of the sequence $\{\bar{X}_n\}$ and (2) the assumptions made about the distributions of r.v.s in the sequence $\{X_i\}$

In particular if $\bar{X}_n \xrightarrow{P} \mu$, we have weak Law of Large Numbers (WLLN) and if $\bar{X}_n \xrightarrow{a.s.} \mu$, we have a Strong Law of Large Numbers (SLLN).

Note 1: More generally, if $E(X_i) = \mu_i, i=1, 2, \dots, n$ and $\bar{\mu}_n = \frac{1}{n} \sum_{i=1}^n \mu_i$, then we explore the convergence of $\bar{X}_n \rightarrow \bar{\mu}_n$

Note 2: Since a.s. Convergence implies convergence in prob so if a SLLN holds for a certain sequence of r.v.s, WLLN will also hold for it. of course, the converse may not be true. weak and Strong LLN are defined in this sense.

Remark: The LLN provides sound footing for the utility of sample mean.

We shall discuss some weak and Strong LLN by changing the factor (2) mentioned above.

Weak Law of Large Numbers (WLLN):

Statement: Let $X_1, X_2, \dots, X_n$ be a sequence of random variables and $\mu_1, \mu_2, \dots, \mu_n$ be their respective expectations and let

$$B_n = \text{Var}(X_1 + X_2 + \dots + X_n) < \infty$$

$$\text{Then } P \left\{ \left| \frac{X_1 + X_2 + \dots + X_n}{n} - \frac{\mu_1 + \mu_2 + \dots + \mu_n}{n} \right| < \epsilon \right\} > 1 - \eta.$$

for all $n > n_0$, where ϵ and η are arbitrary small positive numbers, provided

$$\lim_{n \rightarrow \infty} \frac{B_n}{n^2} \rightarrow 0.$$

Proof: using Chebychev's inequality, to the r.v. $\frac{X_1 + X_2 + \dots + X_n}{n}$, we get for any $\epsilon > 0$,

$$P \left\{ \left| \left(\frac{X_1 + X_2 + \dots + X_n}{n} \right) - E \left(\frac{X_1 + X_2 + \dots + X_n}{n} \right) \right| < \epsilon \right\} > 1 - \frac{B_n}{n^2 \epsilon^2}$$

$$\left[\text{Since } V \left(\frac{X_1 + X_2 + \dots + X_n}{n} \right) = \frac{1}{n^2} V(X_1 + X_2 + \dots + X_n) = \frac{B_n}{n^2} \right]$$

$$\Rightarrow P \left\{ \left| \frac{X_1 + X_2 + \dots + X_n}{n} - \frac{\mu_1 + \mu_2 + \dots + \mu_n}{n} \right| < \epsilon \right\} > 1 - \frac{B_n}{n^2 \epsilon^2}$$

Since ϵ is arbitrary, we assume $\frac{B_n}{n^2 \epsilon^2} \rightarrow 0$ as $n \rightarrow \infty$.

Thus, having chosen two arbitrary small positive numbers ϵ and η , numbers n_0 can be found so that inequality $\frac{B_n}{n^2 \epsilon^2} < \eta$ will hold for $n > n_0$. Consequently, we shall have

$$P \left\{ \left| \frac{X_1 + X_2 + \dots + X_n}{n} - \frac{\mu_1 + \mu_2 + \dots + \mu_n}{n} \right| < \epsilon \right\} > 1 - \eta \quad \forall n > n_0 \quad (\epsilon, \eta).$$

This conclusion leads to the following important results, known as Weak Law of Large numbers.

"With the probability approaching unity or certainty as near as we please, we may expect that the arithmetic mean of values actually assumed by n random variables will differ from the arithmetic mean of their expectations by less than any given number, however small, provided the number of variables can be taken sufficiently large and provided the condition: $\frac{B_n}{n^2} \rightarrow 0$ as $n \rightarrow \infty$ is fulfilled".

Remarks 1. The weak law of large numbers can also be stated as follows: $\bar{X}_n \xrightarrow{P} \bar{\mu}_n$, provided $\frac{B_n}{n^2} \rightarrow 0$ as $n \rightarrow \infty$.

2). For the existence of the law, we assume the following conditions:

i) $E(X_i)$ exists $\forall i$ ii) $B_n = \text{Var}(X_1 + X_2 + \dots + X_n)$ exists and

iii) $B_n/n^2 \rightarrow 0$ as $n \rightarrow \infty$.

Condition (i) is necessary, however conditions (ii) + (iii) are not necessary however (iii) is a sufficient condition.

(iii) WLLN for iid random variables: If the variables $X_1, X_2, \dots, X_n$ are independent and identically distributed i.e. $E(X_i) = \mu$ and $V(X_i) = \sigma^2$ ($\forall i = 1, 2, \dots, n$), then

$$B_n = V(X_1 + X_2 + \dots + X_n) = \sum_{i=1}^n V(X_i) = n\sigma^2 \quad (\text{Covariance terms vanish as r.v.s are independent})$$

$$\text{Hence, } \lim_{n \rightarrow \infty} \frac{B_n}{n^2} = \lim_{n \rightarrow \infty} \frac{\sigma^2}{n} = 0.$$

Thus, the weak law of large number holds for the sequence $\{X_n\}$ of iid r.v.s and we get

$$P\left\{\left|\frac{X_1 + X_2 + \dots + X_n}{n} - \mu\right| < \epsilon\right\} > 1 - \eta, \forall n > n_0.$$

$$\text{i.e. } P\left\{|\bar{X}_n - \mu| < \epsilon\right\} \rightarrow 1 \text{ as } n \rightarrow \infty \Rightarrow P\left\{|\bar{X}_n - \mu| \geq \epsilon\right\} \rightarrow 0 \text{ as } n \rightarrow \infty.$$

Where $\bar{X}_n = \frac{X_1 + X_2 + \dots + X_n}{n}$, this result implies that $\bar{X}_n$ converges in probability to μ i.e. $\bar{X}_n \xrightarrow{P} \mu$.

Note: $E(X_i) = \mu$, $V(X_i) = \sigma^2$, then $E(\bar{X}_n) = \mu$ and $V(\frac{1}{n} \sum_{i=1}^n X_i) = \frac{\sigma^2}{n}$.

Bernoulli's Law of Large Numbers: Let there be a sequence of n Bernoulli trials resulting in a success or failure. If X is the no. of successes in n trials with constant prob p of success for each trial, then $E(X) = np$, $V(X) = npq$, $q = 1 - p$, the variable $\frac{X}{n}$ represents the proportion of successes or the relative frequency of successes, and $E(\frac{X}{n}) = p$ and $V(\frac{X}{n}) = \frac{1}{n^2} V(X) = \frac{pq}{n}$. Then $P\left\{|\frac{X}{n} - p| < \epsilon\right\} \rightarrow 1$ as $n \rightarrow \infty \Rightarrow P\left\{|\frac{X}{n} - p| \geq \epsilon\right\} \rightarrow 0$ as $n \rightarrow \infty$. For any assigned $\epsilon > 0$, this implies that (X/n) converges to p as $n \rightarrow \infty$.

Proof: Applying Chebyshev's inequality to the r.v. $\frac{X}{n}$, we get for any $\epsilon > 0$

$$P\left\{\left|\frac{X}{n} - E\left(\frac{X}{n}\right)\right| \geq \epsilon\right\} \leq \frac{V(X/n)}{\epsilon^2} = \frac{pq}{n\epsilon^2} \leq \frac{1}{4n\epsilon^2}$$

Since the maximum value of pq is at $p = q = \frac{1}{2}$, i.e.

$$\max(pq) = \frac{1}{4}, \text{ i.e. } pq \leq \frac{1}{4}.$$

Since ϵ is arbitrary, we get

$$P\left\{\left|\frac{X}{n} - p\right| \geq \epsilon\right\} \rightarrow 0 \text{ as } n \rightarrow \infty \Rightarrow P\left\{\left|\frac{X}{n} - p\right| < \epsilon\right\} \rightarrow 1 \text{ as } n \rightarrow \infty.$$

Ex. a symmetric die is thrown 600 times. Find the lower bound for the prob. of getting 80 to 120 sixes.

Solⁿ: Let S be the total number of successes, then

$$E(S) = np = 600 \times \frac{1}{6} = 100, \quad V(S) = npq = 600 \times \frac{1}{6} \times \frac{5}{6} = \frac{500}{6}$$

using Chebyshev's inequality we get

$$P\left\{|S - E(S)| < K\sigma\right\} \geq 1 - \frac{1}{K^2}$$

$$\Rightarrow P\left\{|S - 100| < K\sqrt{\frac{500}{6}}\right\} \geq 1 - \frac{1}{K^2}$$

~~Taking $K = 20$~~

$$\therefore P\left\{100 - K\sqrt{\frac{500}{6}} < S < 100 + K\sqrt{\frac{500}{6}}\right\} \geq 1 - \frac{1}{K^2}$$

Taking $K = \frac{20}{\sqrt{500/6}}$, we get $P(80 \leq S \leq 120) \geq 1 - \frac{1}{400 \times \frac{6}{500}} = \frac{19}{24}$

Hence $\boxed{P(80 \leq S \leq 120) = \frac{19}{24}}$

Ex. Use Chebyshev's inequality to determine how many times a fair coin must be tossed in order that the probability will be at least 0.90 that the ratio of the observed number of heads to the number of tosses will lie between 0.4 and 0.6.

Solⁿ : As per Bernoulli's Law of Large Numbers,
we get for any $\epsilon > 0$,

$$P\left\{\left|\frac{X}{n} - p\right| > \epsilon\right\} \leq \frac{1}{4n\epsilon^2} \Rightarrow P\left\{\left|\frac{X}{n} - p\right| < \epsilon\right\} > 1 - \frac{1}{4n\epsilon^2}$$

Since $p = 0.5$ (fair coin i.e. coin is unbiased) and we want the proportion of successes $\frac{X}{n}$ to lie between 0.4 and 0.6, we have

$$\left|\frac{X}{n} - p\right| \leq 0.1 \Rightarrow 0.1 \leq \frac{X}{n} - p < 0.1 \Rightarrow$$

$$p - 0.1 \leq \frac{X}{n} \leq p + 0.1$$

$$\Rightarrow 0.4 \leq \frac{X}{n} \leq 0.6$$

Thus choosing $\epsilon = 0.1$, we have

$$P\left\{\left|\frac{X}{n} - p\right| < 0.1\right\} > 1 - \frac{1}{4n(0.1)^2} = 1 - \frac{1}{0.04n}$$

Since, we want this probability to be 0.9, we fix:

$$1 - \frac{1}{0.04n} = 0.9 \Rightarrow 0.10 = \frac{1}{0.04} \Rightarrow n = \frac{1}{0.10 \times 0.04}$$

$$= 250$$

Hence the required no. of tosses is 250.

Ex. If you wish to estimate the proportion of engineers and scientists who have studied probability theory and you wish your estimate to be correct within 2% with probability 0.95 or more, how large a sample would you take.

- i) If you have no idea what the true proportion is,
- ii) if you are confident that the true proportion is less than 0.2?

Solⁿ : Let $\frac{X}{n} = \hat{p}$, here $\epsilon = 2\% = 0.02$, so that

$$P\left\{\left|\hat{p} - p\right| < 0.02\right\} > 1 - \frac{1}{4n(0.02)^2} = 0.95$$

$$\Rightarrow 0.05 = \frac{1}{4n(0.02)^2} \Rightarrow n = 12,500$$

(ii) By Bernoulli's WLLN $P\left\{\left|\hat{p} - p\right| \leq \epsilon\right\} > 1 - \frac{p(1-p)}{n\epsilon^2}$

Here $p < 0.2 \Rightarrow p(1-p) < 0.16$, Taking $\epsilon = 2\% = \frac{1}{50}$,

$$\text{Hence } P\left\{\left|\hat{p} - p\right| \leq \epsilon\right\} > 1 - \frac{0.16}{n\epsilon^2} = 0.95$$

$$\Rightarrow n = 50 \times 50 \times 20 \times 0.16 = 8000.$$

Ex. For the discrete r.v. X , the prob distn is given as

X	-1	0	1
$P(X)$	$\frac{1}{8}$	$\frac{6}{8}$	$\frac{1}{8}$

(a) Find $P\{|X - \mu_X| \geq 2\sigma_X\}$

(b) Compare this result with that obtained on using Chebychev's inequality.

Soln

(a) $E(X) = -1 \times \frac{1}{8} + 0 \times \frac{6}{8} + 1 \times \frac{1}{8} = 0$

$E(X^2) = 1 \times \frac{1}{8} + 0 \times \frac{6}{8} + 1 \times \frac{1}{8} = \frac{1}{4}$

$V(X) = E(X^2) - (E(X))^2 = \frac{1}{4} \Rightarrow \sigma_X = \frac{1}{2}$

$P\{|X - \mu_X| \geq 2\sigma_X\} = P\{|X| \geq 1\} = 1 - P\{|X| < 1\}$
 $= 1 - P(-1 < X < 1) = 1 - P(X=0)$
 $= 1 - \frac{6}{8} = \frac{2}{8} = \frac{1}{4}$

(b) As per Chebychev's inequality

$P\{|X - \mu_X| \geq 2\sigma_X\} \leq \frac{1}{4}$, In this case both results are same.

Ex. If X is number of scored in a throw of a fair die, show that the Chebychev's inequality gives

$P\{|X - \mu| \geq 2.5\} \leq 0.47$, where μ is the mean of X , while the actual prob. is zero.

Here, X is a random variable which takes the values 1, 2, 3, ..., 6, each with prob $\frac{1}{6}$. Hence $E(X) = \frac{1}{6}(1+2+3+\dots+6) = \frac{1}{6} \times \frac{6 \times 7}{2} = \frac{7}{2}$

$E(X^2) = \frac{1}{6}(1^2+2^2+\dots+6^2) = \frac{1}{6} \times \frac{6 \times 7 \times 13}{6} = \frac{91}{6}$

$V(X) = E(X^2) - (E(X))^2 = \frac{91}{6} - \frac{49}{4} = \frac{35}{12} = 2.9167$

For $K > 0$, Chebychev's inequality gives

$P\{|X - E(X)| \geq K\} \leq \frac{V(X)}{K^2}$

choosing $K = 2.5$, we get $P\{|X - \mu| \geq 2.5\} \leq \frac{2.9167}{6.25} = 0.47$.

The actual prob P is given by

$P = P\{|X - 3.5| \geq 2.5\} = P\{X \text{ lies outside the limits } (3.5 - 2.5, 3.5 + 2.5)\}$
 $= P\{X = 1, 6\}$

But since X is the number on dice when thrown, it can't ~~not~~ lie outside the limits of 1 and 6. In other words $p = P(\phi) = 0$.

Ex. How large a sample must be taken in order that the probability will be at least 0.95 that $\bar{X}_n$ will be within 0.5 of μ (μ is unknown and $\sigma = 1.0$)

Solⁿ: we have $E(\bar{X}_n) = \mu$ and $V(\bar{X}_n) = \frac{\sigma^2}{n}$,

we get for any $c > 0$

$$P\{|\bar{X}_n - E(\bar{X}_n)| < c\} \geq 1 - \frac{\text{Var}(\bar{X}_n)}{c^2}$$

$$\Rightarrow P\{|\bar{X}_n - \mu| < c\} \geq 1 - \frac{\sigma^2}{nc^2} \quad \text{--- (1)}$$

We want n so that $P\{|\bar{X}_n - \mu| < 0.5\} \geq 0.95$

Comparing (1) and (2), we get

$$c = 0.5 = \frac{1}{2} \quad \text{and} \quad 1 - \frac{\sigma^2}{nc^2} = 0.95 \quad \text{and} \quad \sigma = 1 \quad \text{given}$$

$$\therefore 1 - \frac{4}{n} = 0.95 = \frac{4}{n} = 0.05 = \frac{1}{20}$$

$$\Rightarrow n = 80, \quad \text{Hence } n \geq 80$$

CENTRAL LIMIT THEOREM (CLT)

"If X_i ($i = 1, 2, \dots, n$) be the independent random variables such that $E(X_i) = \mu_i$ and $V(X_i) = \sigma_i^2$, then under certain very general conditions, the random variable $S_n = X_1 + X_2 + \dots + X_n$, is asymptotically normal with mean μ and standard deviation σ where

$$\mu = \sum_{i=1}^n \mu_i \quad \text{and} \quad \sigma^2 = \sum_{i=1}^n \sigma_i^2$$

Case 1: De Moivre's Laplace Theorem:

"If $X_i = \begin{cases} 1, & \text{with prob } p \\ 0, & \text{with prob } q \end{cases}$

then the distribution of the random variable $S_n = X_1 + X_2 + \dots + X_n$, where X_i 's are independent, is asymptotically normal as $n \rightarrow \infty$.

$$\sim N(\mu, \sigma^2)$$

⇒ Lindberg-Lévy Theorem (For i.i.d. r.v.s)
 Case (ii) If $X_1, X_2, \dots, X_n$ are independently and identically distributed random variables with

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$$\left. \begin{aligned} E(X_i) &= \mu_1 \\ V(X_i) &= \sigma_1^2 \end{aligned} \right\}, i = 1, 2, \dots, n.$$

then the sum $S_n = X_1 + X_2 + \dots + X_n$ is asymptotically normal with mean $\mu = n\mu_1$ and variance $\sigma^2 = n\sigma_1^2$.

⇒ Liapounoff's Central Limit Theorem:

Let $X_1, X_2, \dots, X_n$ be independent r.v.s such that

$$\left. \begin{aligned} E(X_i) &= \mu_i \\ V(X_i) &= \sigma_i^2 \end{aligned} \right\}; i = 1, 2, \dots, n.$$

Let us suppose that third absolute moment say ρ_i^3 of X_i about its mean exists i.e. $\rho_i^3 = E(|X_i - \mu_i|^3)$; $i = 1, 2, \dots, n$ is finite

Further let
$$\rho^3 = \sum_{i=1}^n \rho_i^3,$$

if $\lim_{n \rightarrow \infty} \frac{\rho}{\sigma} = 0$, then the sum $X = X_1 + X_2 + \dots + X_n$ is asymptotically normal $N(\mu, \sigma^2)$, where $\mu = \sum_{i=1}^n \mu_i$ and $\sigma^2 = \sum_{i=1}^n \sigma_i^2$.

Remark: If the variables X_i ($i = 1, 2, \dots, n$) are identical, then

$$\rho^3 = \sum_{i=1}^n \rho_i^3 = n\rho_1^3 \text{ and } \sigma^2 = \sum_{i=1}^n \sigma_i^2 = n\sigma_1^2$$

$$\therefore \frac{\rho}{\sigma} = \frac{n^{\frac{1}{3}} \rho_1}{n^{\frac{1}{2}} \sigma_1} = \frac{\rho_1}{\sigma_1} \cdot \frac{1}{n^{\frac{1}{6}}} \rightarrow 0 \text{ as } n \rightarrow \infty.$$

Thus for identical variables, the condition of Liapounoff's theorem is satisfied.